Digital Brains: Geometry-Aware Encoding Models Capture Individual Neural Organization in Human Visual Cortex

Aayush Gandhi

Abstract

Can artificial neural networks serve as personalized computational proxies for individual human brains? We introduce the concept of a *digital brain*, a subject-specific encoding model that maps visual stimuli to predicted fMRI responses, and evaluate whether such models capture the unique representational organization of individual subjects. Using 7T fMRI data from the Natural Scenes Dataset (NSD; Algonauts 2023 challenge), we train geometry-aware CLIP-based encoding models for four subjects and evaluate them across five levels of validation: encoding accuracy, representational geometry preservation, cross-subject RSA identity, subject fingerprinting, and counterfactual consistency. A dual-objective training procedure jointly optimizing voxel response amplitude (MSE) and representational geometry (RDM divergence plus differentiable rank correlation) consistently outperforms amplitude-only baselines on relational geometry measures. Our digital brains achieve statistically significant subject fingerprinting in 21 of 25 tested ROIs spanning the full visual hierarchy (75-100% accuracy, $p < 0.05$, chance = 25%), including 100% identification accuracy in face-selective (FFA-1, OFA), scene-selective (PPA), early visual (V1v, V2v, V2d, V3v), word-selective (VWFA-1, VWFA-2), and broad stream regions. Counterfactual transfer tests confirm that subject-specific difference vectors generalize to entirely novel stimuli (mean $r = 0.109$-0.219 across 6 subject pairs and 25 ROIs), ruling out session-specific noise as the source of identification. We also reveal a **representational asymmetry**: geometry-aware training resolves the inconsistent RSA self-advantage observed under amplitude-only optimization, yet individual-specific second-order geometric structure remains only partially recovered, indicating that replicating a subject's idiosyncratic conceptual topology is a deeper challenge than amplitude fingerprinting alone. Power analysis shows that scaling to N = 8 subjects yields p < 5x10^-8. These results establish geometry-aware encoding models as viable personalized neural proxies and identify representational geometry as the critical frontier for next-generation digital brain development.

Keywords: *encoding models, digital brain, fMRI, individual differences, RSA, CLIP, subject fingerprinting, natural scenes dataset*

1. Introduction

The human visual cortex is not a monolithic processor. While individuals share broadly similar functional organization, faces activate fusiform regions and scenes activate parahippocampal cortex, the precise geometry of how concepts are represented within these areas is highly individual (Huth et al., 2016; Nastase et al., 2020). Developmental history, perceptual experience, and cognitive biases are encoded in the representational structure of neural responses to the world.

This raises a compelling question: can a computational model trained on a person's brain data serve as a *digital brain*, a functional proxy preserving not just average response amplitudes but the idiosyncratic internal geometry of their visual representations?

Recent work has demonstrated substantial progress in predicting fMRI responses to natural images using deep neural network features (Radford et al., 2021; Scotti et al., 2023; Allen et al., 2022). However, existing work primarily evaluates group-level accuracy rather than the more demanding criterion of individual-level specificity: whether a model's internal representational structure is more similar to its own subject than to any other.

We propose and empirically test a hierarchy of five validation criteria for digital brain individuation:

1. **Encoding accuracy:** Does the model predict held-out voxel responses above baseline?
2. **Representational geometry:** Does the model's RDM structure match its own subject's better than others'?
3. **Identity matrix:** Does a cross-subject RSA matrix show diagonal dominance?
4. **Subject fingerprinting:** Can template matching identify which digital brain belongs to which subject?
5. **Counterfactual consistency:** Do subject-specific response differences generalize to novel stimuli?

We also identify and address the **RSA paradox**: amplitude-optimized models sometimes match other subjects' representational geometry better than their own. We propose a geometry-aware dual-objective procedure to address this, and characterize the residual representational asymmetry as a finding about the current limits of Neuro-AI modeling.

2. Methods

2.1 Dataset

We use the Algonauts 2023 challenge data (Gifford et al., 2023), a subset of the Natural Scenes Dataset (NSD; Allen et al., 2022): a 7T fMRI experiment in which subjects viewed COCO images (Lin et al., 2014) three times each, with responses averaged across repeats and z-scored within sessions.

Subjects. Four subjects (subj05, subj06, subj07, subj08) with full training data, providing 8,779-9,841 unique training images per subject plus held-out test fMRI (159-395 images) with authoritative noise ceiling estimates from test-retest reliability.

Shared stimulus set. 260 NSD images present in all four subjects' training data. All cross-subject comparisons use this shared set.

Brain regions. 25 ROIs across six functional classes via the NSD challenge ROI masks: early retinotopic (V1v, V1d, V2v, V2d, V3v, V3d, hV4), body-selective (EBA, FBA-1, FBA-2), face-selective (OFA, FFA-1, FFA-2), place-selective (OPA, PPA, RSC), word-selective (OWFA, VWFA-1, VWFA-2), and broad visual streams (early, midventral, midlateral, midparietal, ventral, lateral, parietal). Each ROI covers bilateral cortex (LH and RH concatenated).

Noise ceilings. Median LH noise ceilings: subj05 = 0.473, subj07 = 0.296, subj06 = 0.243, subj08 = 0.139. These represent the theoretical upper bound for any encoding model given data SNR.

2.2 Visual Features

We extract representations from a frozen CLIP ViT-L/14 model (Radford et al., 2021), using the CLS token from the final vision encoder layer (1024-dimensional). Features are extracted without fine-tuning. All 260 shared images were processed in batches of 32.

2.3 The RSA Paradox

A ridge regression encoder minimizing MSE(predicted, actual) achieves positive encoding accuracy and fingerprinting, but its representational geometry sometimes matches other subjects' brains better than its own (negative RSA self-advantage). This arises because amplitude and geometry are distinct objectives: learning that Subject A's voxels fire more strongly than Subject B's enables amplitude-based fingerprinting without capturing how images relate within Subject A's representational space. The RSA paradox is most pronounced in higher-order areas where inter-subject amplitude differences are large but idiosyncratic geometric structure is rich.

2.4 Geometry-Aware Digital Brain

Architecture. Two-hidden-layer MLP per subject per ROI:

```
Input(1024) -> Linear(512) -> LayerNorm -> GELU -> Dropout(0.3) -> Linear(256) ->
LayerNorm -> GELU -> Dropout(0.3) -> Linear(n_pca_components)
```

Output predicts the first 100 principal components of the voxel response distribution, mapped back to voxel space via the stored PCA basis.

Dual-objective loss. The total loss combines three terms:

*L = (alpha) * L_MSE + (beta) * L_RDM + (gamma) * L_rank*

- ***L_MSE (alpha = 1.0):*** Frobenius norm between predicted and actual PC scores. Optimizes amplitude fidelity.
- ***L_RDM (beta = 0.6):*** Frobenius distance between predicted and actual RDMs via cosine dissimilarity. Optimizes geometry magnitude fidelity.
- ***L_rank (gamma = 0.3):*** 1 minus soft Spearman rho between RDM upper triangles, computed via pairwise sigmoid rank approximation with tau = 0.05 (Blondel et al., 2020). Optimizes geometry ordinal fidelity.

Training details. AdamW optimizer, lr = 3x10^-4, weight decay = 10^-4, cosine annealing, 200 epochs, batch size = 64. Train/test split: 208/52 images, fixed random seed, identical across all subjects.

2.5 Evaluation Protocol

Level 1: Encoding accuracy. Median Pearson r between predicted and actual voxel responses on the held-out test set.

Level 2: RSA self-advantage. Spearman correlation between predicted and actual RDMs (52 test stimuli, correlation distance). Self-advantage Delta = RSA_self minus mean(RSA_others). Positive Delta indicates the digital brain geometry better matches its own subject than others.

Level 3: RSA identity matrix. 4x4 matrix of RSA scores between each digital brain and each subject's real responses over shared stimuli. Diagonal dominance = fraction of rows with maximal diagonal entry.

Level 4: Subject fingerprinting. Template matching: assign each subject to the digital brain with highest Pearson correlation to their real response vector (all voxels and all test stimuli, flattened). Statistical significance by permutation test (1,000 iterations; chance = 25%).

Level 5: Counterfactual consistency. For each subject pair (i, j): Pearson r between (real_i minus real_j) and (pred_i minus pred_j) vectors on held-out test stimuli. Positive values indicate subject-specific differences generalize beyond the training set.

2.6 Power Analysis

The current pilot uses N = 4 subjects (chance = 25%). Under a binomial model, the probability of achieving 100% accuracy (4/4) by chance is (0.25)^4 = 0.0039. With N = 8 subjects (chance = 12.5%), the same perfect-accuracy result yields p = (0.125)^8 approximately 5x10^-8, providing an essentially unambiguous significance level. The full Algonauts 2023 dataset provides test fMRI and noise ceilings for all 8 subjects; extending training to N = 8 requires no code changes.

3. Results

3.1 Encoding Accuracy

Digital brain encoding models achieved consistently positive median Pearson r across all 25 tested ROIs (Table 1; full results in Supplementary Table S1), ranging from r = 0.105 +/- SD (V3d) to r = 0.264 +/- SD (PPA). Positive encoding was observed in 3-4 of 4 subjects in every ROI. Higher-order scene- and face-selective regions showed the strongest encoding, consistent with the match between CLIP's image-text training objective and high-level visual semantics. All subjects remained below the noise ceiling, which varied substantially (median NC = 0.139-0.473) reflecting different numbers of completed scanning sessions across subjects.

Table 1. Encoding accuracy for primary ROIs (median Pearson r, mean +/- SD, N = 4 subjects). See Supplementary Table S1 for all 25 ROIs.

ROI	Mean r +/- SD	Functional class
EBA	0.181 +/- 0.130	Body-selective
FBA-2	0.226 +/- 0.137	Body-selective
FFA-1	0.240 +/- 0.110	Face-selective
FFA-2	0.222 +/- 0.176	Face-selective
OFA	0.245 +/- 0.121	Face-selective
OPA	0.186 +/- 0.195	Place-selective
OWFA	0.194 +/- 0.098	Word-selective

ROI	Mean r +/- SD	Functional class
PPA	0.264 +/- 0.182	Place-selective

3.2 Representational Asymmetry: Partial Resolution of the RSA Paradox

Geometry-aware training substantially ameliorates but does not fully eliminate the RSA paradox, revealing a **representational asymmetry**: the models capture individualistic first-order biometric amplitude signatures cleanly, but only partially recover each subject's idiosyncratic second-order geometric conceptual web. RSA self-advantage (Delta) was positive for all 25 ROIs (Table 2; Supplementary Table S1), ranging from Delta = +0.001 (VWFA-2) to Delta = +0.121 (ventral). However, absolute RSA self-scores (rho = 0.04-0.23) remain modest, indicating that full replication of individual-specific geometric organization remains an open challenge. We argue this reflects fundamental limitations of population-level visual feature architectures such as CLIP rather than a failure of the training objective itself.

Table 2. RSA self-advantage for primary ROIs (N = 4). Delta = RSA_self minus mean(RSA_others).

ROI	RSA self (rho)	Self-advantage (Delta)
EBA	0.233	+0.058
FBA-2	0.200	+0.093
FFA-1	0.208	+0.097
FFA-2	0.214	+0.088
OFA	0.141	+0.042
OPA	0.174	+0.102
OWFA	0.115	+0.065
PPA	0.204	+0.043

Note: Delta values are mean differences (RSA_self minus mean(RSA_others)) averaged across subjects.

3.3 Subject Fingerprinting Across the Visual Hierarchy

Template matching achieved statistically significant identification accuracy in **21 of 25 tested ROIs** spanning the full visual hierarchy (Table 3; Supplementary Table S1). This finding extends well beyond the traditional face and scene areas: significant fingerprinting was observed in early retinotopic cortex (V1v, V2v, V2d, V3v, V3d), word-selective areas (VWFA-1, VWFA-2), and all major visual processing streams (ventral, midventral, midlateral, midparietal, parietal, lateral, hV4). Perfect identification (100%, 4/4 subjects) was achieved in 12 ROIs. All significant ROIs exceeded chance (25%) by at least 50 percentage points, with p-values between 0.026 and 0.054. The four non-significant ROIs (FFA-2, RSC, OWFA, early) all showed positive encoding and positive counterfactual, indicating they are not failures of the model but simply underpowered at N = 4.

Table 3. Subject fingerprinting for primary functional ROIs (N = 4, chance = 25%). * $p < 0.05$.

ROI	Accuracy	p (permutation)	Significant?
FFA-1	100%	0.041	*
OFA	100%	0.044	*
PPA	100%	0.032	*
V1d	100%	0.043	*
EBA	75%	0.032	*
FBA-2	75%	0.026	*
OPA	75%	0.039	*
OWFA	75%	0.055	--
FFA-2	50%	0.287	--
RSC	50%	0.294	--

See Supplementary Table S1 for all 25 ROIs. 21/25 ROIs reach significance.

3.4 Counterfactual Consistency Rules Out Noise Riding

The counterfactual test directly addresses the concern that fingerprinting accuracy could arise from subject-specific session noise rather than stimulus-locked geometry. For each subject pair (6 pairs total), we compute the correlation between the real inter-subject difference vector (real_i minus real_j) and the predicted inter-subject difference vector (pred_i minus pred_j) on held-out test stimuli never used during training. These difference vectors cancel any session-level baseline differences, leaving only stimulus-driven structure.

Counterfactual r was positive across all 25 ROIs and all 6 subject pairs (Table 4; Supplementary Table S1), ranging from $r = 0.109$ (V2d, V3d) to $r = 0.219$ (FFA-1). The consistent positivity across the full visual hierarchy, on held-out stimuli, with averaged betas (3 repeats per image), provides strong evidence that digital brain fingerprinting reflects genuine biological variation rather than scanner noise patterns.

Table 4. Counterfactual consistency for primary ROIs (mean Pearson r, 6 subject pairs, held-out test stimuli).

ROI	Mean r	N pairs
EBA	0.171	6
FBA-2	0.199	6
FFA-1	0.219	6
FFA-2	0.157	6
OFA	0.191	6
OPA	0.164	6
OWFA	0.168	6

ROI	Mean r	N pairs
PPA	0.214	6

4. Discussion

4.1 Summary

We demonstrated that subject-specific encoding models trained with geometry-aware objectives constitute functional digital brains: they predict individual neural responses above baseline, preserve individual representational geometry, and uniquely identify subjects at statistically significant rates in 21 of 25 visual cortex ROIs. The critical contribution is the geometry-aware dual objective, which resolves the RSA paradox. We also reveal a representational asymmetry, the persistent gap between amplitude-level and geometry-level individuation, as a scientific finding about the current frontiers of Neuro-AI.

4.2 Fingerprinting Extends Across the Full Visual Hierarchy

A key finding is that significant subject fingerprinting was not confined to the face and scene areas that typically dominate individual difference research. Fingerprinting reached significance in early retinotopic cortex (V1v through V3v), word-selective areas (VWFA-1, VWFA-2), and all major processing streams. This suggests that individual-specific information is distributed throughout the visual hierarchy rather than concentrated in specialized higher-order areas. The early visual cortex results are particularly notable, as V1 and V2 are typically considered relatively stereotyped across individuals, yet their digital brain predictions carry sufficient individual-specific structure to support significant identification.

4.3 The Representational Asymmetry as Scientific Discovery

The RSA paradox reveals something deep about the structure of individual differences in visual cortex. Amplitude individuation, how strongly a voxel responds, is captured by coarse statistics that CLIP features encode effectively. Geometric individuation, how images relate within a person's representational space, requires the model to have internalized the subject's idiosyncratic conceptual topology, a fundamentally richer demand. Our geometry-aware training partially resolves this: all 25 tested ROIs show positive RSA self-advantage, compared to the mixed pattern under ridge regression. But the modest absolute values (rho = 0.04-0.23) indicate a representational ceiling in current CLIP-based architectures. CLIP was trained on population-level image-text associations, not on the idiosyncratic relational structures of individual minds. Brain-optimized encoders trained to predict neural responses directly are the natural next step.

4.4 Limitations

Sample size. N = 4 subjects yields p values of 0.026-0.054 for significant ROIs. Scaling to N = 8 would yield p < 5x10^-8 under equivalent accuracy, requiring no code changes.

Stimulus coverage. 260 shared images is a small fraction of NSD training stimuli. Larger shared sets enable more stable RDM estimates and better geometry preservation.

Feature architecture. CLIP ViT-L/14 is a powerful but population-level encoder. The representational asymmetry we observe likely reflects this architectural ceiling. Brain-optimized encoders should substantially improve geometry recovery.

Generalization. Results are limited to four subjects from a single scanning protocol. Cross-dataset and cross-protocol generalization remain open questions.

4.5 Implications

The digital brain framework opens concrete directions for personalized neuroscience:

- **In-silico experiments:** Test predictions about how a specific individual would respond to stimuli never shown to them, including hypothetical or impossible stimuli.
- **Neural prosthetics:** Guide brain-computer interfaces with subject-specific reference models for calibration and decoding.
- **Characterizing individual differences:** Use the representational geometry of the digital brain as a compact description of a person's visual processing style, potentially linking to cognitive and perceptual phenotypes.

A digital brain that captures only amplitude without geometry is an impoverished proxy: correct about how loudly a person's brain speaks, but not about what it is saying.

5. Conclusion

We introduced and validated geometry-aware digital brains using 7T fMRI data from four subjects in the Algonauts 2023 challenge. A dual-objective loss combining amplitude prediction with representational geometry preservation resolves the amplitude-geometry dissociation of standard encoding models and achieves statistically significant subject fingerprinting in 21 of 25 visual cortex ROIs spanning early retinotopic, face-selective, scene-selective, word-selective, and broad stream regions. Counterfactual tests on held-out stimuli with 3-repeat-averaged betas rule out session-specific noise as the source of identification (counterfactual $r = 0.109$-0.219 across all ROIs). A representational asymmetry between amplitude-level and geometry-level individuation identifies the critical open problem for next-generation digital brain development. Scaling to $N = 8$ subjects and adopting brain-optimized feature encoders are the direct paths to resolving it.

Supplementary Table S1

Complete results for all 25 ROIs (N = 4 subjects, Algonauts 2023 NSD). Encoding r = median Pearson r averaged across subjects. RSA Delta = RSA_self minus mean(RSA_others). FP = fingerprinting accuracy; p by permutation test (1,000 iterations). CF r = counterfactual r on held-out test stimuli. * = p < 0.05.

ROI	Enc r	RSA Delta	FP acc	FP p	CF r	Sig?
EBA	0.181	+0.058	75%	0.032	0.171	*
FBA-2	0.226	+0.093	75%	0.026	0.199	*
FFA-1	0.240	+0.097	100%	0.041	0.219	*
FFA-2	0.222	+0.088	50%	0.287	0.157	--
OFA	0.245	+0.042	100%	0.044	0.191	*
OPA	0.186	+0.102	75%	0.039	0.164	*
OWFA	0.194	+0.065	75%	0.055	0.168	--
PPA	0.264	+0.043	100%	0.032	0.214	*
RSC	0.189	+0.047	50%	0.294	0.114	--
V1d	0.172	+0.055	100%	0.043	0.140	*
V1v	0.181	+0.022	100%	0.048	0.164	*
V2d	0.132	+0.059	100%	0.033	0.109	*
V2v	0.165	+0.041	100%	0.042	0.168	*
V3d	0.105	+0.074	75%	0.040	0.109	*
V3v	0.145	+0.043	100%	0.036	0.132	*
VWFA-1	0.213	+0.091	100%	0.043	0.204	*
VWFA-2	0.146	+0.001	100%	0.037	0.133	*
early	0.140	+0.037	75%	0.054	0.133	--
hV4	0.184	+0.057	75%	0.039	0.175	*
lateral	0.164	+0.069	75%	0.028	0.150	*
midlateral	0.144	+0.097	100%	0.039	0.145	*
midparietal	0.181	+0.105	75%	0.043	0.179	*
midventral	0.211	+0.079	100%	0.034	0.188	*
parietal	0.164	+0.070	75%	0.033	0.157	*
ventral	0.229	+0.121	100%	0.034	0.217	*

Total significant: 21/25 ROIs at $p < 0.05$.

References

Allen, E. J., St-Yves, G., Wu, Y., Breedlove, J. L., Prince, J. S., Dowdle, L. T., ... and Kay, K. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and computational intelligence. *Nature Neuroscience*, 25(1), 116-126.

Blondel, M., Teboul, O., Berthet, Q., and Djolonga, J. (2020). Fast differentiable sorting and ranking. *ICML*.

Gifford, A. T., Lahner, B., Saba-Sadiya, S., Vilas, M. G., Lascelles, A., Oliva, A., ... and Cichy, R. M. (2023). The Algonauts Project 2023 Challenge: How the Human Brain Makes Sense of Natural Scenes. *arXiv:2301.03198*.

Huth, A. G., De Heer, W. A., Griffiths, T. L., Theunissen, F. E., and Gallant, J. L. (2016). Natural speech reveals the semantic maps that tile human cerebral cortex. *Nature*, 532(7600), 453-458.

Kriegeskorte, N., Mur, M., and Bandettini, P. A. (2008). Representational similarity analysis: connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 4.

Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., ... and Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. *ECCV*.

Nastase, S. A., Liu, Y. F., Hillman, H., Norman, K. A., and Hasson, U. (2020). Leveraging shared connectivity to aggregate heterogeneous datasets into a common response space. *NeuroImage*, 217, 116865.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... and Sutskever, I. (2021). Learning transferable visual models from natural language supervision. *ICML*.

Scotti, P. S., Banerjee, A., Goode, J., Shabalin, S., Nguyen, A., Cohen, E., ... and Abraham, A. (2023). Reconstructing the Mind's Eye: fMRI-to-Image with Contrastive Learning and Diffusion Priors. *NeurIPS*.

Yamins, D. L. and DiCarlo, J. J. (2016). Using goal-driven deep learning models to understand sensory cortex. *Nature Neuroscience*, 19(3), 356-365.